Proof of Work VS Proof of Stake

Mining capacity depends on computational power

Validating capacity depends on the stake in the network

Miners receive block rewards to solve a cryptographic puzzle

Validators do not receive a block reward, instead, they collect transaction fees as reward

Hackers would need to have a computer powerful than 51% of the network to add a malicious block, leading to 51% attack

Hacker would need to own 51% of all the cryptocurrency on the network, which is practically impossible and therefore, making 51% attacks impossible.

🔹 Validation Method (Block kaise verify hota hai)

PoW (Proof of Work):

- Computer **hard calculation / puzzle solve** karta hai
- Isko hi **mining** bolte hain

PoS (Proof of Stake):

- Jitne **coins tum stake (lock)** karte ho
- Utna hi **verification ka control** milta hai

👉 **Simple:**

PoW = mehnat

PoS = coins lock

🔹 Consensus Algorithm (Validator kaun banega?)

PoW:

- Jo **sabse pehle puzzle solve** kare
- Wahi **validator** ban jaata hai

PoS:

- Ek **algorithm choose karta hai validator**
- Choice depend karti hai **kitne coins stake kiye hain**

👉 **Simple:**

PoW = fastest computer

PoS = zyada stake wala

🔹 Rewards (Reward kaun kaise kamata hai?)

PoW:

- Jo block **pehle mine** kare
- Usko **block reward** milta hai

PoS:

- Validator ko **network fees** reward me milti hain

👉 **Simple:**

PoW = mining reward

PoS = fees reward

🔹 Efficiency (Kitna efficient hai?)

PoW:

- Zyada power + zyada bijli lagti hai ⚡
- Isliye less efficient

PoS:

- Mining machine ki zarurat nahi
- Isliye more efficient ⏳

👉 Simple:

PoW = bijli khata hai

PoS = bijli bachata hai

🔹 Centralization (Control kiske paas ja sakta hai?)

PoW:

- Jinke paas bahut powerful machines hoti hain
- Wo network pe dominate kar sakte hain

PoS:

- Jo early aaye aur bahut zyada coins ho
- Unke paas power aa sakti hai

👉 Simple:

PoW = powerful miners ka risk

PoS = ameer validators ka risk

🔹 Equipment (Kaunsi machine chahiye?)

PoW:

- ASICs, GPUs jaise heavy machines chahiye

PoS:

- Normal server / laptop bhi kaam kar sakta hai

👉 Simple:

PoW = mehngi machine

PoS = simple machine

🔹 Cryptocurrency Examples

PoW coins:

- Bitcoin (BTC)
- Litecoin (LTC)

PoS coins:

- Ethereum 2.0 (ETH)
- Binance Coin (BNB)

🧠 One-line yaad rakhne ka formula

- 🔨 PoW = Mehnat + Machine
- 💰 PoS = Coins + Stake

👉 Ethereum (ETH) ab PoS pe kaam karta hai

- Pehle Ethereum **PoW (Proof of Work)** pe tha
- **2022 me "The Merge" ke baad** Ethereum **PoS (Proof of Stake)** pe shift ho gaya

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🧠 Easy samajhne ke liye:

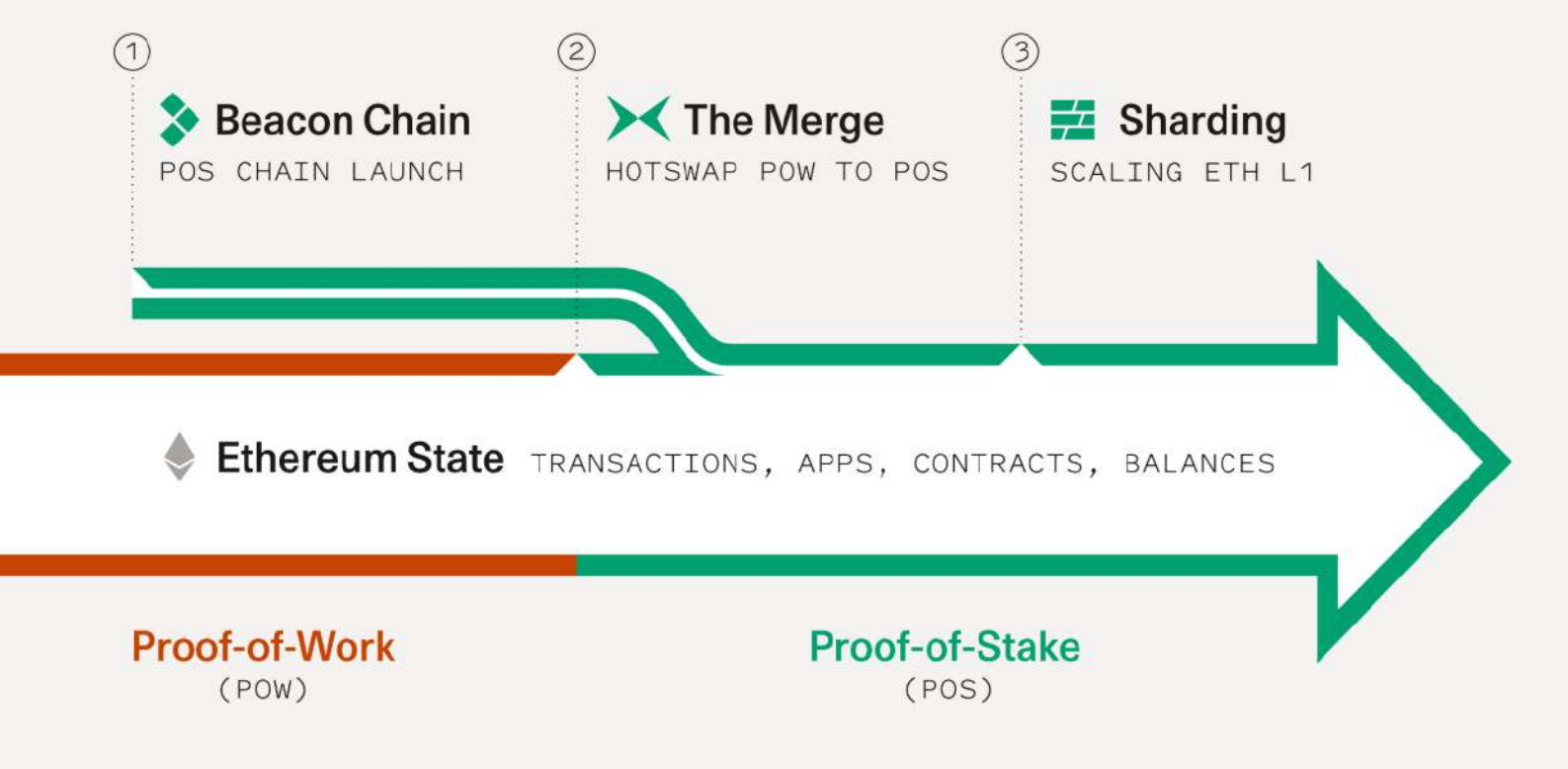
- **Pehele (PoW):**
  - Mining hoti thi
  - GPU se puzzle solve
  - Zyada electricity ⚡
- **Ab (PoS):**
  - Mining ❌
  - Validators ETH **stake (lock)** karte hain
  - Kam electricity 🌱

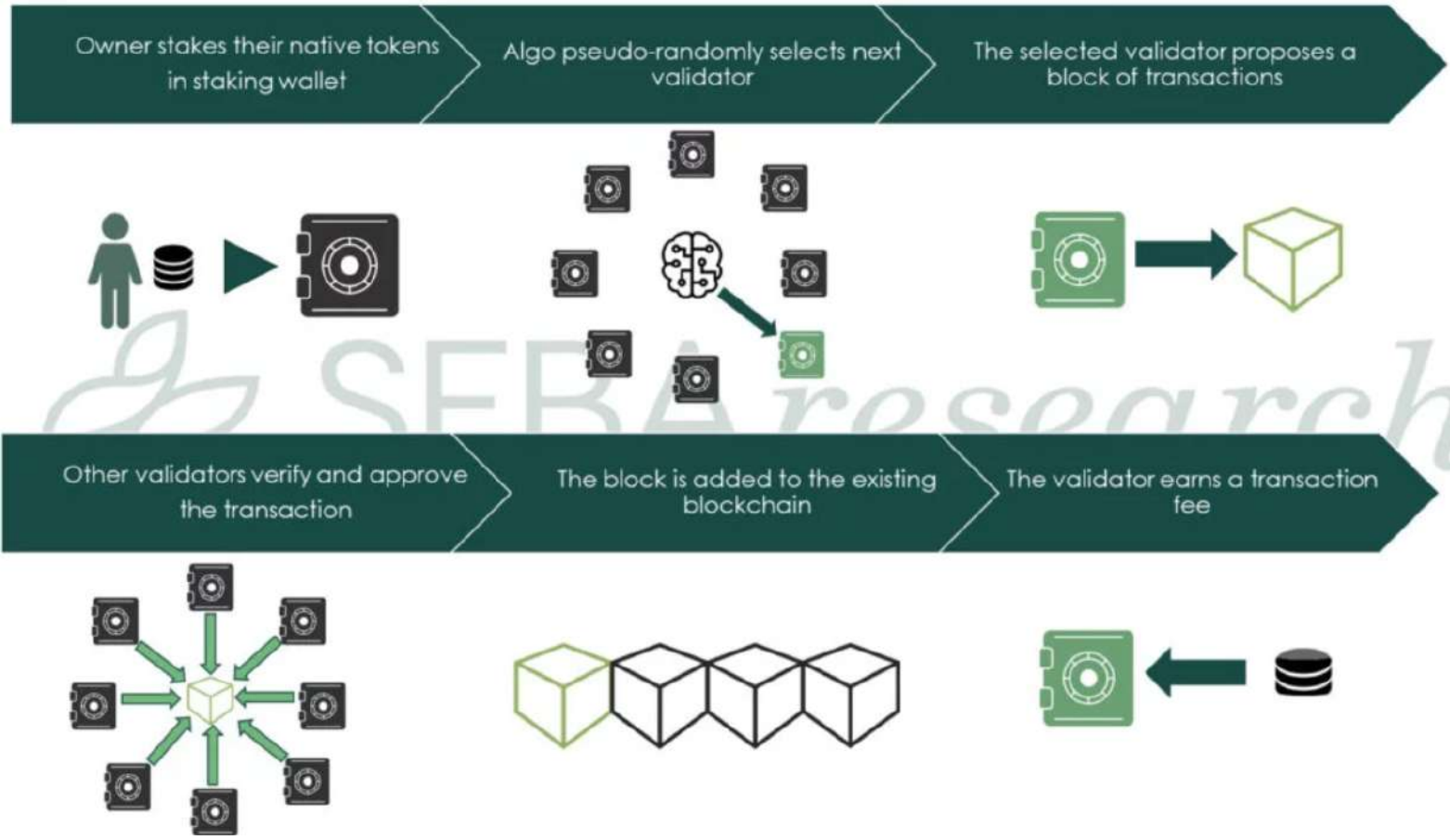
🔐 Validator banne ke liye:

- **32 ETH stake** karna padta hai
- Ya phir **staking pool** ke through bhi kar sakte ho

📝 One-line yaad rakhne ka trick:

> **Bitcoin = PoW**
> **Ethereum = PoS**


1
Beacon Chain
POS CHAIN LAUNCH
2
The Merge
HOTSWAP POW TO POS
3
Sharding
SCALING ETH L1
Ethereum State
TRANSACTIONS, APPS, CONTRACTS, BALANCES
Proof-of-Work
(POW)
Proof-of-Stake
(POS)



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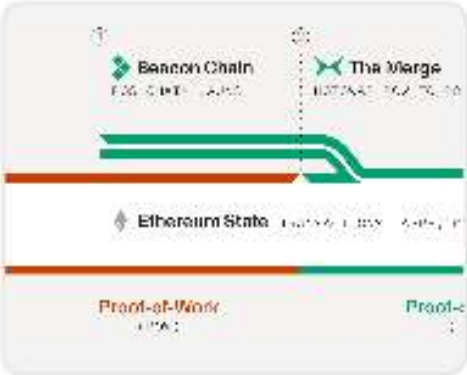
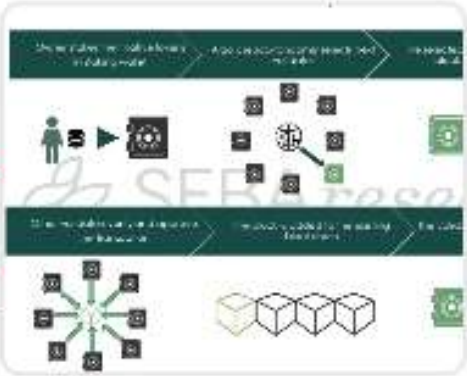
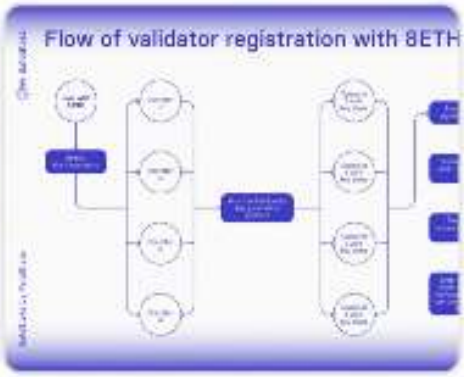
🔄 The Merge kya tha? (Simple)







- Ethereum **pehle PoW** pe chal raha tha (mining hoti thi)
- **2022 me Ethereum ne PoW chhod diya**
- Aur **PoS (Proof of Stake)** pe shift ho gaya
- Is change ko **The Merge** bola jata hai

👉 **Simple line:**

Mining band ❌, Staking start ✅

🧠 Ab Ethereum PoS pe kaise kaam karta hai?

🔹 Step 1: ETH stake karo

- User apne **ETH lock (stake)** karta hai
- Minimum **32 ETH** chahiye (direct validator banne ke liye)

🔹 Step 2: Validator bano

- Jisne ETH stake kiya → **Validator**
- Validator ka kaam:
  - Transactions check karna
  - New block propose karna

🔹 Step 3: Network choose karta hai

- Ethereum ka system **random + stake ke basis** pe
- Kisi ek validator ko bolta hai:

> "Tum next block banao"

🔹 Step 4: Reward ya Penalty

- Agar **sahi kaam kiya** → ETH reward 💰
- Agar **galat / cheating** → ETH cut ❌
  (isko **slashing** bolte hain)

💸 ETH ka reward kaise milta hai?

- Mining reward ❌
- Staking reward + transaction fees ✅

⚡ PoS kyun better hai Ethereum ke liye?

- 🔥 Electricity bahut kam
- 🚀 Fast & scalable
- 🌱 Environment friendly
- 🛡️ Secure (cheating me paisa jata hai)

🧠 One-glance comparison (yaad rakhne ke liye)

- **Old Ethereum:** PoW → Mining → GPU
- **New Ethereum:** PoS → Staking → Validator

📌 Final one-line yaad rakhna

> Ethereum = PoS (stake ETH → validator → reward)